Smart AI Fan

TECHNOMANCERS (T-2026); atomQUEST 2026 Stage 1; Track 1 Fan

A Team & Institutional Context

Team: TECHNOMANCERS (T-2026); **Institute:** Aditya Institute of Technology and Management, Tekkali; **Department:** Electronics and Communication Engineering.

Members (5): Harsha Vardhan Malipeddi, Dabbiru Rochan Kumar, Simhadri Sri Teja, Guluvindala Ganesh Manikanta, Nakka Ugandhar. **Corresponding author:** H. V. Malipeddi (harshavardhanmalipeddi1@gmail.com).

B Problem & Idea Framing

Track and areas of exploration: Track 1 Fan, combining three areas: (1) fan as smart-home enabler (temperature/humidity-triggered devices); (2) sensor-based multi-directional swing (pedestal-fan area, adapted to a ceiling fan); (3) blade design for energy efficiency. **Why combined:** one sensing-and-control system drives automation and swing, while the blades set the operating point both act on.

Idea/mechanism: a 1,400 mm-sweep ceiling fan. A local controller reads temperature/humidity for comfort automation and device triggers. A self-locking worm pan/tilt head steers the whole airflow disc, with rotor stopped before tilt and a supervisor-enforced ceiling-height check (full ±20° needs ≥ 2.660 m ceiling; level-only at 2.650–2.660 m). Quasi-isotropic [0/±45/90]s carbon/epoxy blades run on a 24 V sensorless-FOC BLDC drive, under a locked-firmware supervisor MCU that can cut actuator power on any anomaly.

Baseline vs target: no measured baseline of ours exists. Versus a fixed-speed, fixed-direction, manual fan: steerable airflow (fixed → ±20° tilt); automated response ≤ 5 s with ≥ 95 % success over 10–20 scenarios; a device-trigger path; defined safe fallback on sensor/network loss. Air delivery ≥ 245 m^3/min and service value ≥ 4.1 m^3/min/W are IS 374 targets; ≈ 8.7 m^3/min/W is indicative (245 ÷ 28.3 W), not measured.

C Technical Feasibility

Sensors feed comfort automation and closed-loop FOC speed control (≤ 0.5 s). Plausibility gates (−10…+80 °C, ≤ 3 K/s) route implausible input to a safe default curve; a hardware NTC trips at 165 °C. The self-locking worm holds position on power loss, backed by encoders, limit switches and supervisor power-cut: three independent defenses. **Key specifications (paper values):** 380 RPM top, tip speed 27.86 m/s under the 30 m/s cap; rotor energy ≈ 89.5 J; blade 165 g; worm ≥ 6/2.5 N·m; anchor demand 139.9 N vs 1,500 N ultimate; cable 6.5 kN; electrical per IS 302-2-80 (compliance not established). 456 RPM is a hazard case, never a target. **Open items:** moving-mass roll-up (7.00 vs 6.80 kg) and thin containment margin (128.8 J hazard-case energy vs 130 J rating) are unresolved.

D Execution Feasibility

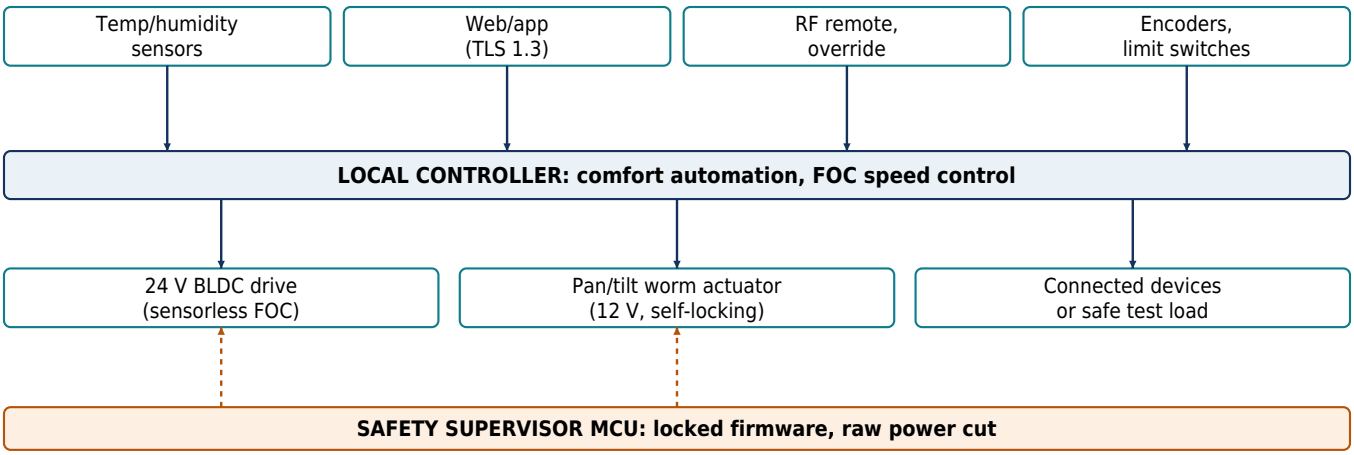

Figure 1. Block diagram; control priority: supervisor, local, RF, web, automation.

Components: custom composite rotor, 24 V FOC BLDC motor and controller, supervisor and local MCUs, temperature/humidity sensors, custom-rated worm actuators with encoders/limit switches, 24 V SMPS, optional Wi-Fi/BLE, RF remote, device trigger or safe test load.

Rough cost/BOM (order of magnitude, unquoted): Round 2 Level 1 prototype ≈ INR 10,000 (7,000–14,000); full-fan prototype ≈ INR 25,000 (17,000–33,000). **Sourcing risk:** custom CFRP blades (tooling, cure, verification; no supplier datasheet yet) and custom-rated worm actuators.

Build plan to Round 2 (Level 1 prototype: low-voltage control demonstrator): (1) sensing and comfort control; (2) verify override precedence and sensor-disconnect/network-loss fallback; (3) device trigger on a safe test load; (4) record results against predefined acceptance tests; (5) motor/rotor integration only after safety review and approval.

Status: paper design; no hardware built, nothing measured. Powered-rotor testing and certification follow safety review. **Evidence repository** (calculations, BOM, risk register): github.com/harsha-vardhan-2006/atomquest